LOUDOUN ROBOTICS · STONE HILL MIDDLE SCHOOL

What makes a robot a robot?

Before we build one — what are we actually building?

Twelve Wednesdays · Sep 16 – Dec 9

WHY ROBOTICS

We build robots to go where we cannot.



Mars

No air, no rescue. Rovers have worked there for years with nobody nearby.



Inside a reactor

Radiation that would kill a person in minutes. A robot just goes in.



Inside a person

Surgical robots move steadier than any human hand, through a smaller cut.



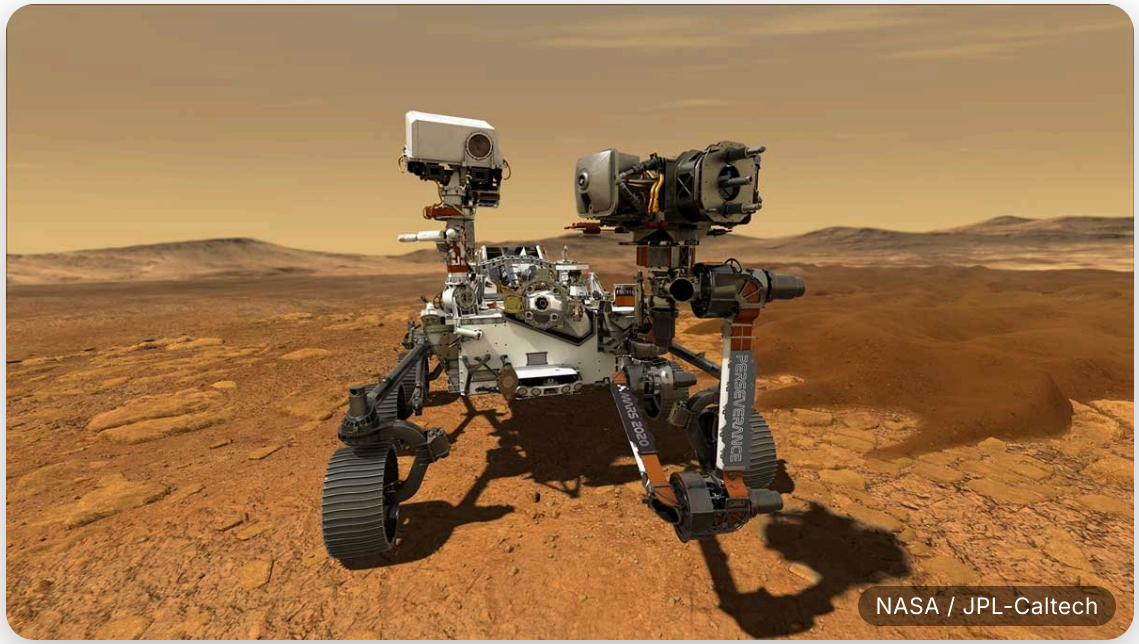
4 km down

Pressure that would crush a submarine. We map the seafloor this way.

Every one of them had to be **taught what to do** by somebody.

WHY THEY CANNOT JUST BE REMOTE CONTROLLED

Mars is too far away to drive.



NASA / JPL-Caltech

3–22 min

for a command to reach Mars — and just as long for the answer to come back

Do the maths. You see a cliff in a photo and send “STOP.” That photo left Mars up to 22 minutes ago, and your command needs another 22 to get there. By the time the rover hears you, it drove off the cliff **three quarters of an hour** earlier.

So nobody drives it. **It has to decide for itself.**

WHICH MEANS IT NEEDS THREE THINGS

Sense. Think. Act. By itself.



That loop is the whole definition. **Miss any one and it is a machine, not a robot.**

THE LINE

Who is doing the thinking?

Toaster

NO

Acts on a timer. Senses nothing, decides nothing.

RC car

NO

Senses and acts — but **you** do the thinking.

Robot vacuum

YES

Finds a wall, decides to turn, drives on. Nobody watching.

Mars rover

YES

Has to be. Help is 22 minutes away.

A phone-controlled car is a *tool* — your brain is the missing piece, and you have to be there. **Take the thinking out of a person and put it inside the machine, and it becomes a robot.** That is the whole job.

WHY IT MATTERS FOR YOU

The robot is not really the point.

You are going to be wrong a lot in here. On purpose. That is the part worth having.

1. **Say what you think will happen** — before you find out. Out loud, in writing, where everyone can see it.
2. **Test it properly.** Three runs, one thing changed at a time. Otherwise you learn nothing.
3. **Be wrong without flinching.** A surprising result is the interesting one. It means something you believed was not true.
4. **Find out why, and fix it.** Then defend your choice with the numbers — not with "it looked better."

Those four work on **anything**. The LEGO is just what we practise them on.

YOUR TWELVE WEEKS

You will build the loop one piece at a time.

ACT

Motors turn wheels. You will make a robot drive an exact distance, turn a real 90°, and choose gears for speed or pushing power.

Weeks 1 – 6

SENSE

Sensors arrive, one at a time. Your robot stops being blind — it starts noticing the world instead of just moving through it.

Weeks 7 – 9

THINK

You decide what the robot does with what it senses — and in what order. At the edge, back off. On contact, push.

Weeks 10 – 12

This is the order you **learn** them — not the order they **run**. By Week 10 all three run together, sense first, in one loop, many times a second.

QUESTION TWO

So what is it made of?

THE BRAIN

EV3 brick

A small computer. It holds your program and tells everything else what to do.

HOW IT ACTS

Motors

Turn electricity into movement. Everything your robot *does* in the ring, it does with these.

HOW IT SENSES

Sensors

Touch, color and ultrasonic. They report numbers — they do not decide anything.

THE BODY

Beams, pins, gears

The frame. It has to hold together when another robot hits it.

THE ENERGY

Battery

A robot with a flat battery is a paperweight. Charge it, every week.

THE INVISIBLE PART

Your program

You cannot hold it or drop it. Without it, nothing above does anything at all.

HOW IT SENSES

Three sensors. Three different questions.

Touch

“Am I being pressed — yes or no?”
The simplest answer a robot can get. Perfect for knowing you have found your opponent.

Week 7 · port 1

Color

“How bright is the floor?”
Pointed down, it tells the ring floor from the tape edge — so you can stop before you drive out.

Week 8 · port 3

Ultrasonic

“How far away is that?”
Sends out a sound and times the echo — like a bat. Gives you centimeters.

You have this one · port 4

A sensor never tells you *what* something is. Ultrasonic says “30 cm away” — it cannot tell a robot from a wall. **Deciding what that means is your job.**

WHERE THINGS PLUG IN

Letters move. Numbers sense.

LETTERS — MOTORS	
A	Your first robot's motor (Weeks 1–2)
B	Left drive wheel (Week 4 on)
C	Right drive wheel (Week 4 on)
D	Spare

NUMBERS — SENSORS	
1	Touch (Week 7)
2	Spare
3	Color, facing down (Week 8)
4	Ultrasonic, facing forward (you have this)

Week 4 is the day this changes. Your drive motors move from A to B and C — so a program written in Week 1 will not drive the new robot. You rebuild it. That catches everyone; now it will not catch you.

THE INVISIBLE PART

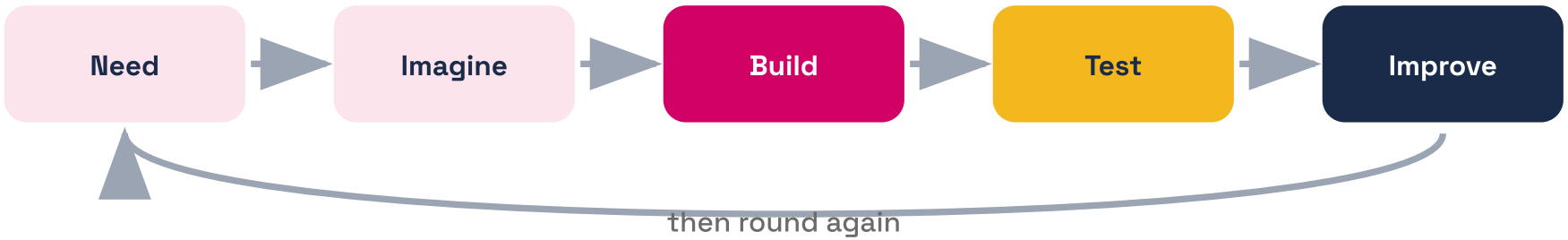
The robot does exactly what you wrote.

Not what you meant. Not what you wanted. Exactly what you wrote, in exactly the order you wrote it — every single time.

> When your robot does something odd, it is not being difficult and it is not broken. It is following your instructions perfectly. **The bug is in the instructions** — and finding it is the actual skill.

This is why we **predict first**, then measure. The gap between the two is where the learning is.

Nobody gets it right first time.



A surprising result is a clue, not a failure. Your notebook is where you keep the evidence — predict first, keep the numbers, sketch what changed.

DECEMBER 9

Where this is going.

A sumo tournament, in front of your families. Two robots in a ring. Nobody touching the controls.

Your robot will have to **find** its opponent, **stay** inside the ring, and **decide** what to do when both happen at once — on its own, in about a second.

> You will also be asked to explain *why* you built it that way. “We measured, and this was better” beats “it looked cool.”

Next: open your notebook to **page 6** and build something that drives.

WEEK 1 · WEDNESDAY SEP 16

Build and drive

Tonight: Build Bevel Bot. Make it drive and stop.

0:00
Welcome
Find your team, check the kit, open the notebook.

5 min

0:05
Build
Bevel Bot from LEGO's pictures. Take turns. A mentor checks wheels, gears, pins and cables before anything runs.

25 min

0:30
Program & test
Your first program: motor A, drive, stop. Measure travel from the same front edge every time.

12 min

0:42
Evidence
Write the distance down, and one thing you noticed.

3 min

0:45
Clean up
Programs saved and named, kit counted, notebooks away.

5 min

0:50
Spare
Late starts, flat batteries, things that need rebuilding.

10 min

THE WORDS FOR TONIGHT

Electric motor — turns electrical energy into motion. **Program** — a set of instructions a computer follows. **Sequence** — the order those instructions run in.

PREDICT FIRST — BEFORE ANYTHING MOVES

Where will the first run stop?

DONE TONIGHT

Bevel Bot built, your own A-motor program, one measured drive and stop.

IF TIME REMAINS

Three 100 cm trials.

WEEK 2 · WEDNESDAY SEP 23

Find your grip

Tonight: Find the setup that slides at the steepest angle.

0:00
Arrive & set up

Ramp, angle gauge and catch area checked by a mentor.

5 min

0:05
Key idea & predict

Friction and bracing. Everyone writes a guess before the board is raised.

7 min

0:12
Build & test

Raise the board slowly. Record the angle at first slide, three times per setup. Change one thing only — the tyre or the surface.

25 min

0:37
Evidence & share

Work out mean and range, then compare with another team. Do you agree?

8 min

0:45
Pack up

Bevel Bots stay built — you need them again next week.

5 min

0:50
Spare

Resets, tries that did not count, rebuilding.

10 min

THE WORDS FOR TONIGHT

Friction — the force that opposes sliding between two surfaces in contact. **Brace** — a support that helps a structure resist changing shape.

PREDICT FIRST — BEFORE ANYTHING MOVES

Which tyre or surface will slide at a steeper angle?

DONE TONIGHT

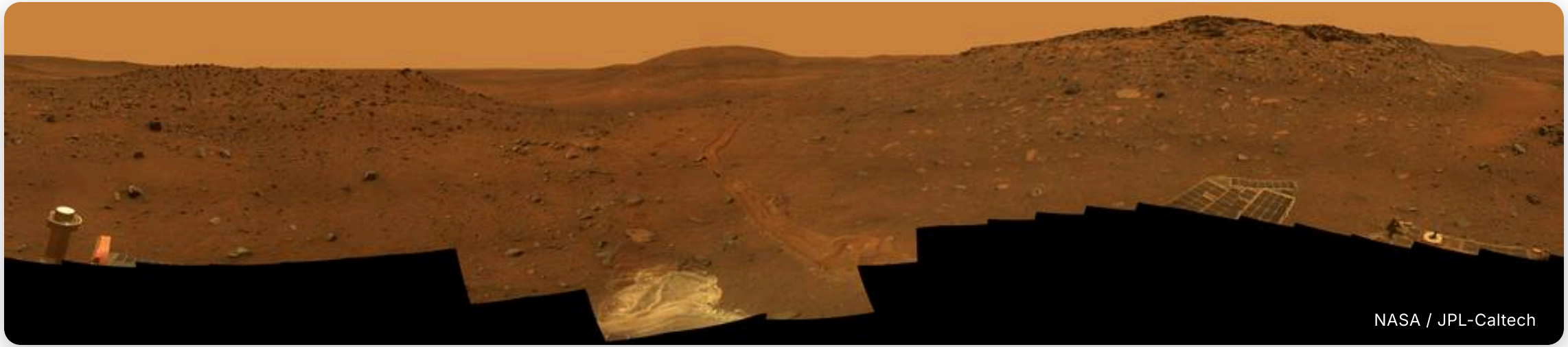
Two grip setups, three valid slides each, one brace comparison.

IF TIME REMAINS

Powered pushing demo · predict a third setup · does added weight change the angle?

WEEK 2 · WHY THIS MATTERS

NASA lost a rover to this.



APRIL 2009

Driving backwards, Spirit’s wheels broke through a hard crust. Underneath was loose, slippery sand.

EVERY ATTEMPT

Each try at driving out dug it in deeper. The more they tried, the worse it got.

IT NEVER MOVED AGAIN

Built to last 90 days, it had already worked for five years. The mission ended here.

Not the motors. Not the cold. Not the computer. **It was grip.**

So they buried one on Earth.



A full-size copy of the rover, a tray of soil mixed to behave like the sand on Mars — and engineers deliberately driving it in to get stuck the same way.

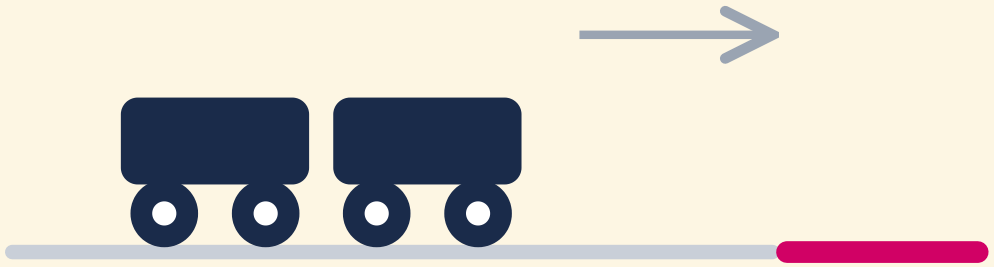
> They tried escape moves here for **months** before sending a single command to Mars. Watching the wheels in this photo are Colette Lohr and Kim Lichtenberg, while Paolo Bellutta enters the next driving command.

Remember the 22 minutes? **You cannot take a command back** — so you test it somewhere you can.

The same thing decides December 9.

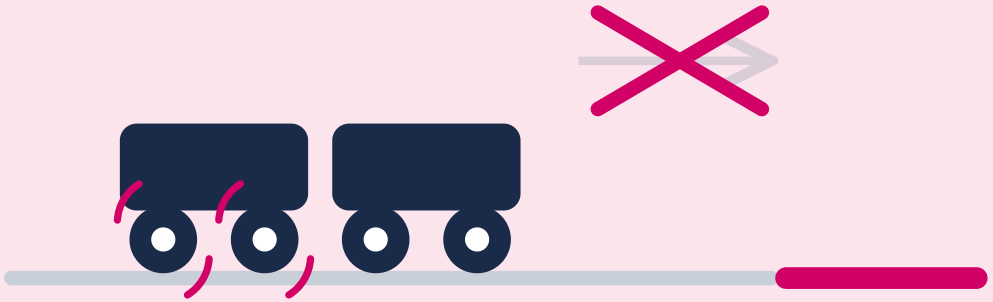
Tyres hold

Force reaches the floor. Your opponent goes backwards, towards the tape.



Tyres slip

Motors at full power, wheels spinning, nobody moving. All of it wasted.



THE THING WORTH REMEMBERING

Your motor does not decide how hard you can push. **Your tyres do.** Grip sets the ceiling — next week's gears only decide where you sit underneath it.

Three rules, or tonight measures nothing.

Change one thing

Same tyres when you compare surfaces. Same surface when you compare tyres. Change two and you will never know which one did it.

Mass · position · board all fixed

A roll is not a result

If it rolls or tips instead of sliding, that try **does not count**. Write it down, fix the setup, run it again — never average it in.

Reset flat every time

Three trials, then stop

Three tells you whether your setup is steady. A fourth tells you almost nothing. Spend the time on a harder question instead.

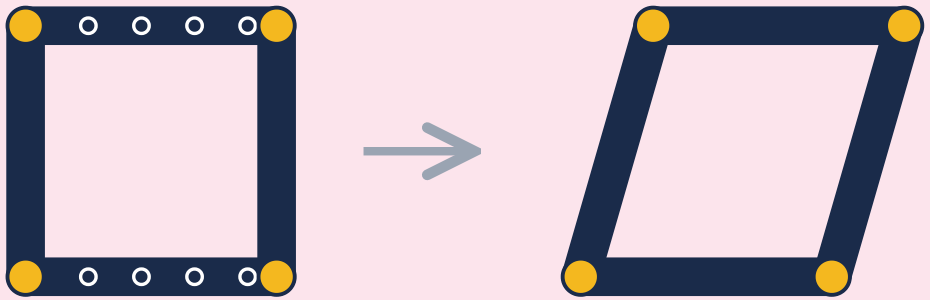
Then mean and range

Three trials are **clues, not proof** — but only if the setup stayed the same.

A rectangle is not rigid. A triangle is.

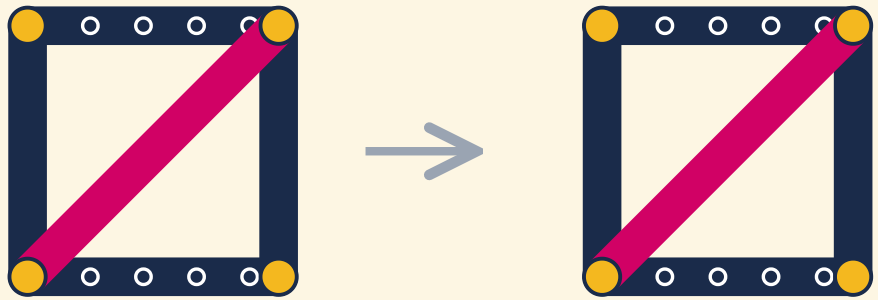
No brace

One pin per corner — and every pin is a hinge. The same push squashes it into a parallelogram.



One diagonal

Now it is two triangles. To squash it, the diagonal would have to stretch — and beams do not stretch.

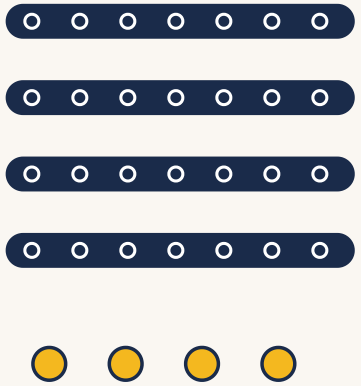


Nothing broke in the first one. **It just changed shape** — which is exactly what loses a sumo match.

Build one to pass around.

1 · THE PARTS

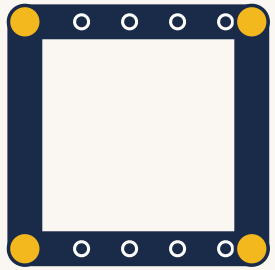
Four beams, four pins



All four beams the same length.

2 · PIN THE CORNERS

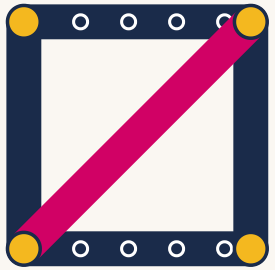
Push it — it squashes



Let them feel it give before you explain anything.

3 · ADD THE DIAGONAL

Same push — it holds



Then send them to find the same pattern on Bevel Bot.

- **Use the loose pins, not the friction ones.** Smooth pins pivot freely, so the squash is obvious.
- **Bigger racks better.** A small tight rectangle barely moves.
- **The diagonal must be longer than the sides** — about 1.4×. Check you have a long enough beam.
- **Pass it round first.** Feel it squash, then feel it refuse.

Three LEGO challenges. One of them fits.

BEST FIT

Robot Bulldozer

Builds a sturdy front blade, real contact with an opponent, and grip while pushing.

A small attachment on the robot they already have.

Cargo Carrier

Explores weight placement and secure mounting — both affect traction and balance.

Useful later. Building the tray takes time they do not have tonight.

Rescue Tow Truck

Explores traction under load and strong connections.

Less direct — towing links are not something sumo needs.

Bulldozer wins because the blade becomes a **sumo pusher** — and that is tonight's grip and bracing, used for something.

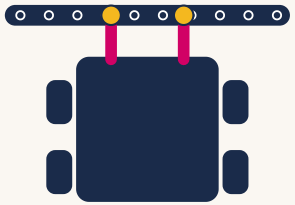
WEEK 2 · GOING FURTHER

Build your sumo pusher.

Your robot can drive. **But can it move an opponent?**

Build

A horizontal beam across the front, clear of the floor and the wheels. **Two separated mounting points** — one will just pivot.



Test

Push a small, lightly loaded cardboard box along a marked 30 cm lane.

Observe

Do the wheels slip? Does the bar twist? Does the box slide sideways?

Improve

Change **one** feature — bar width, height, or bracing — and repeat.

SUCCESS LOOKS LIKE

The box crosses the line on three trials and the attachment never loosens. Same box, same surface, same motor setting. **Stop straight away if the motor stalls.**

IN THE NOTEBOOK

“What limited our pushing ability, and what evidence shows our change helped?”

WEEK 3 · WEDNESDAY SEP 30

Speed or shove?

Tonight: Compare gear setups. Pick your trade-off.

0:00
Arrive & set up
Lane marked, wall or board set for the ultrasonic sensor.

5 min

0:05
Key idea & predict
Gears, torque, gear ratio. Write your guess before anything turns.

5 min

0:10
Build the gear base
LEGO's gearbox model. Some parts are already built. Take turns, and ask a mentor to check the gears before you run it.

25 min

0:35
Swap gears & time it
Three runs on one gear pair. Swap to the other. Three more. **Change nothing else.**

10 min

0:45
Evidence & share
Mean and range for each pair. Which was faster, and what did it cost?

5 min

0:50
Spare
Rebuilds, re-runs, gears that will not sit together.

10 min

THE WORDS FOR TONIGHT

Gear — a wheel with teeth that lock into another one, to pass the turning along. **Torque** — how hard something twists. **Gear ratio** — driven teeth ÷ driving teeth.

PREDICT FIRST — BEFORE ANYTHING MOVES

Which will be fastest? Which will pull best?

DONE TONIGHT

Gear base built, two gear pairs tested, three runs each.

IF TIME REMAINS

A third gear pair · work out the speed · a mentor-led tug test.

Fast, or strong? You only get one.

Fast

Gets across the ring first. Reaches the edge before anyone can stop it. But it can be **shoved aside**.

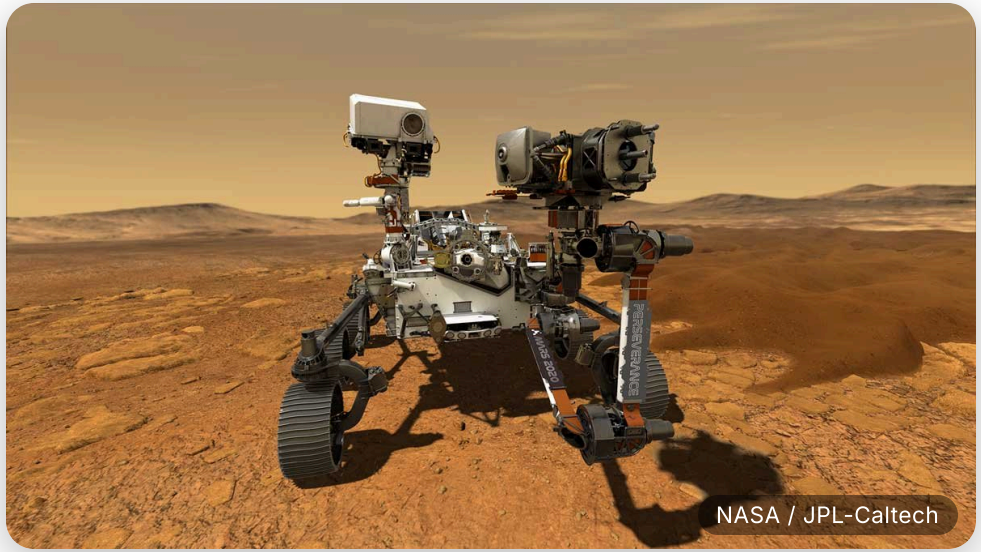
or

Strong

Pushes anything out of its way. Almost impossible to move. But it is **slow getting there**.

Your robot has **one motor**, and it cannot try harder. So what decides which kind of robot you have? **Someone else already faced this exact choice.**

152 meters per hour.



NASA / JPL-Caltech

That is the top speed of NASA's newest Mars rover. Not a typo — **a hundred and fifty-two meters in an hour.**

- ABOUT 33× SLOWER THAN YOU WALK
- ~40 MINUTES TO CROSS A FOOTBALL FIELD
- UNDER 200 WATTS — A CAR USES 150,000

Engineers who can land on Mars built something slower than walking. **Why would they do that?**

WEEK 3 · THE ANSWER

Because it can climb a rock as tall as your knee.

40 cm

The height of rock it will drive straight over. It clears obstacles as big as its own wheel — 52.5 cm.

Knee-high, to you

45°

The tilt it is built to survive without going over. Drivers keep it under 30° anyway.

Your ramp, last week

6 motors

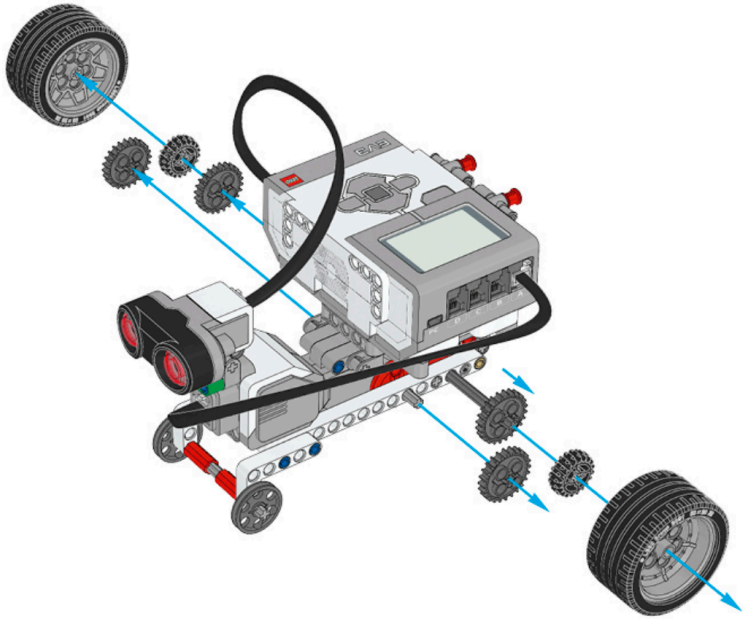
One per wheel, plus four more just for steering. All of it geared for pulling, none of it for racing.

Torque over speed

TONIGHT'S TRADE

NASA gave up speed and bought climbing with it. **You cannot have both** — gears let you choose which one you get, and tonight you measure the cost.

The gears come off. That is the point.



Building instructions © LEGO® Education · MINDSTORMS EV3 Core Set 45544

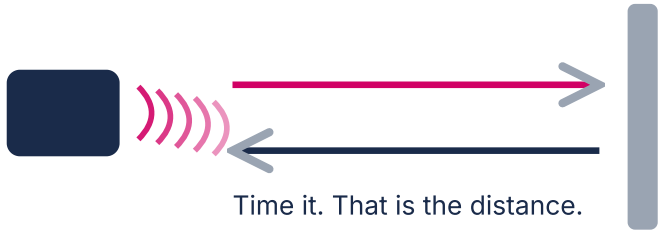
One motor. A gear train to each wheel. The gears slide off the axles — so you can **put different ones on and run it again.**

> Everything else stays bolted together. Same brick, same motor, same wheels, same weight. **The gears are the only thing that changes** — which is what makes your two numbers worth comparing.

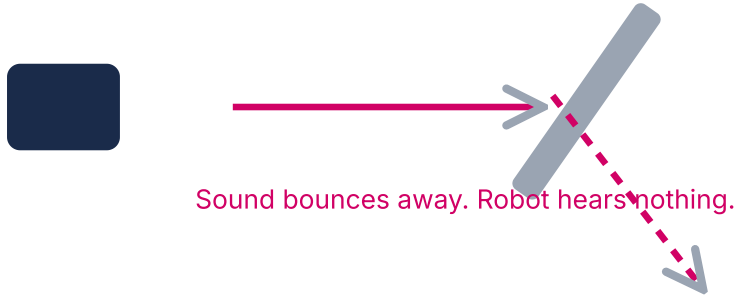
Build it once, then change one thing. **But how will you measure what changed?**

You don’t. The robot does it for you.

Flat on: the echo comes back



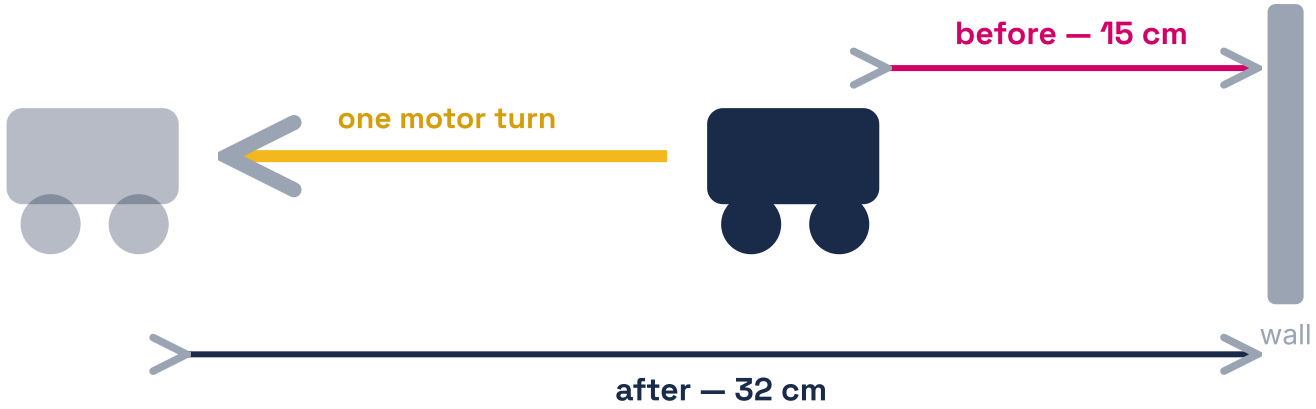
At an angle: it never comes back



WHY THAT MATTERS

That black unit with two round eyes sits on the front of your gear base. It shouts a sound at the wall. Then it **times how long the echo takes to come back**. No tape measure. No chasing it across the floor. **Tonight it is your ruler** — and in December the same sensor is how your robot finds its opponent.

Press one button. The robot does the rest.



1 · PARK IT

At least **10 cm** from the wall, sensor pointing straight at it.

2 · PRESS

The center button. It reads the wall and **remembers** the number.

3 · ONE TURN

The motor turns **exactly once**, then stops. Same every run.

4 · READ IT

It measures again. The **difference** is how far you moved.

Speed is distance ÷ time. That is all it is.

1 The formula you already know

speed = distance ÷ time

2 The sensor gives centimeters. The motor gives degrees.

Centimeters are fine as they are — or **÷ 100** for **meters**, the unit scientists use.

Degrees need **÷ 360** to become **whole turns**.

3 Three numbers, one formula

SPEED · CM/S
cm ÷ seconds

SAME SPEED · M/S
(cm ÷ 100) ÷ s

MOTOR · RPS
(deg ÷ 360) ÷ s

TRY IT — 30 CM IN 4 SECONDS

30 ÷ 4 = **7.5 cm/s** · 0.3 ÷ 4 = **0.075 m/s** — the same speed, written two ways.

WEEK 3 · BEFORE YOU GO

So — which will you build?

You started tonight with a question you could only guess at. Now you have **numbers**.

You measured

How far one motor turn carries each gear pair, three times over, with nothing else changed.

You found the cost

Whatever you gained in one direction, you paid for in the other. That is the trade.

Now you choose

This is the evidence you pick your gearing with.

REMEMBER WHAT NASA PICKED

They chose **slow** — 152 meters an hour — because on Mars, getting stuck is worse than arriving late. **Your ring is not Mars.** What does it reward?

Four things a distance reading can mean.

A small number

Something is close. But is it your opponent, or **just a wall?**

A big number, or nothing

Nothing is close. **Do not charge yet** — you could be aiming at empty floor.

An error on the screen

Something is unplugged or broken. **Stop and check it.**

A number that looks fine but is wrong

Your program cannot tell. This is the one that beats you. Try angled walls, stuff in the background, and another robot nearby.

THE HABIT

A number on the screen is **a clue, not a fact.** There is a session later where the motors stay switched off, so you can find out what the sensor really knows **before** you let it drive.

Last week the robot gave you a number. Was it right?

Things to try tonight. **None of them need the laptop.** You already have the gear base, and the brick shows you the sensor on its own screen.

How it works

It listens for an echo. That is all it does.

How far to trust it

Twenty readings without moving. They will not all match.

What it cannot see

Where the beam goes, and what sends it back.

PORT VIEW

Press the **center button**, pick **Port View**, choose your sensor’s port. The number is live. **No program needed.**

WEEK 4 · FIVE MINUTES

The sound goes there *and back*.

1. **Put the sensor 30 cm from a wall. Use the tape.**
2. **Before you look at the screen — how far did the sound go?**

   30 out + 30 back = 60 cm
3. **Now look. The screen says 30.**

SO THE SENSOR DIVIDES BY 2

60 ÷ 2 = **30 cm** · and your tape measure already said 30

Nothing moved. The number changed.

Put the robot and a box down. **Do not touch either one.** The tape says exactly 30 cm. Now read the screen 20 times and write down every number.

29.4 29.6 29.5 29.7 29.4 29.5 29.6 29.5 …

They jump around

Up a bit, down a bit, no pattern. **Taking an average fixes this.**

They are all a bit low

The tape says 30. Every reading is under. **An average will not fix that.**

WHY THEY ARE ALL LOW

Check **where you measured from.** Your tape started at the front bumper. The sensor sits a bit further back, so it is reading from a different place. **Every number is short by that gap — and it always will be.**

WEEK 4 · TEN MINUTES · PAPER

How many tries is enough?

You have 20 readings. Add up the first 3 and divide by 3. Then do all 10. Then all 20. **Write the three answers in a row.**

First 3	All 10	All 20
29.5	29.5	29.5

SO HOW MANY DID YOU NEED?

3 was enough. The other 17 moved the answer by less than a millimeter.

How far do you go in one wheel turn?

1. **Measure your wheel across the middle. It is 56 mm.**
2. **Multiply by π. That is how far it rolls in one turn.**

   π × 56 mm = 17.6 cm
3. **Now roll it one turn and measure. Do you get 17.6?**

> Now **lift the wheels off the ground** and run it again. The motor still says one full turn. You went **nowhere**. That is slipping taken as far as it goes — and the motor cannot tell the difference.

IF YOU MEASURE THE WHEEL WRONG

Say 54 mm instead of 56 → you are off by 0.6 cm every turn. **After 10 turns you are 6 cm out.**

WEEK 4 · FIFTEEN MINUTES · TWO TEAMS

Guess what is inside. Without opening it.

One team swaps the gears **where the other team cannot see.** The other team gets one motor turn and a tape measure.

1 · Measure

One motor turn moved it **5.9 cm**.

2 · Divide

5.9 ÷ 17.6 = **1/3 of a wheel turn**.

3 · Flip it

The motor turned 3 times for every 1 wheel turn. That is **3:1**.

4 · Check

Open it and count the teeth. **Were you right?**

WHAT YOU JUST DID

You worked out what was inside a closed box **without opening it** — just from how it moved. That is how scientists work out what is inside the sun.

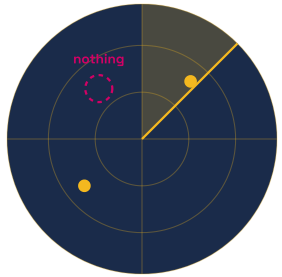
WHERE THIS GOES

There are only two ways to hide from a sensor.

An F-22 is **62 feet long**. On radar it comes back about the size of a **marble**. It does that two ways — and **you will do both tonight.**



U.S. Air Force photo · Master Sgt. Andy Dunaway · public domain



the screen that should be showing it

1 · Point the echo somewhere else

Flat panels at hard angles. The beam arrives and leaves **sideways**, away from whoever sent it.
Tonight: **a board you turn.**

2 · Swallow the echo

Radar-absorbing paint takes the energy in and turns it into a little **heat** instead of sending it back.
Tonight: **a cushion.**

WATCH How radar detection in America’s stealthiest jet works youtube.com/watch?v=AVgddLxN4f0

STILL AN ARGUMENT

Nobody agrees which sensor a car should use.

This is not a solved question that you are being taught the answer to. **It is being argued about right now, with a great deal of money on it.**

Cameras only

Tesla took the radar out in 2021 and the ultrasonic sensors out in 2022. Their argument: people drive with two eyes, so **light should be enough.**

Several at once

Almost everyone else carries lidar, radar and cameras together. Their argument: **each one fails at something different**, so stack them until the gaps close.

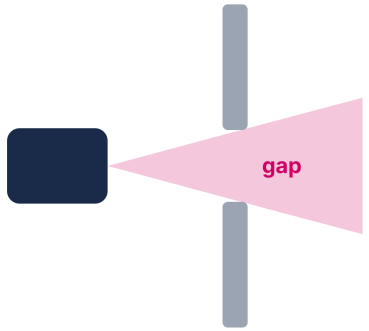
YOU ALREADY KNOW HOW TO THINK ABOUT THIS

It is the same question you asked about the cushion. **What are you trying to see, and what is it made of?** A dark car at dusk, fog, a white truck against a bright sky — each one is something that a particular sensor cannot see.

youtube.com/watch?v=b_4qMNyg1nI · youtube.com/watch?v=Kcq0tjmvGOs

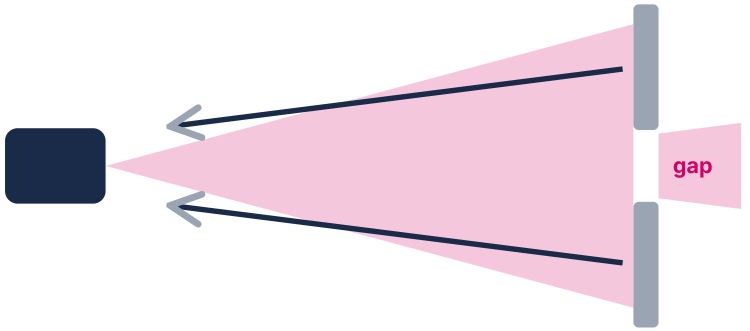
The beam spreads out. That is why the gap vanishes.

Close to the wall — it finds the gap



Beam covers **2.7 cm** — it all fits through the gap.

20 inches away — the gap disappears



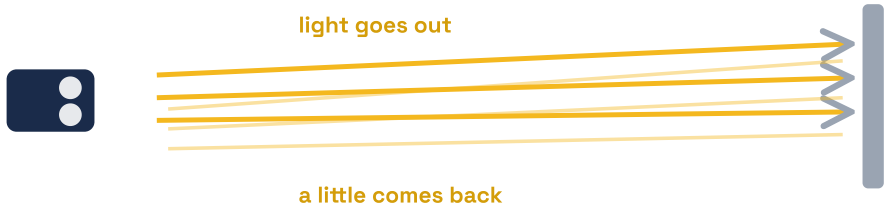
Beam covers **27 cm**. The gap is 18% of it. The wall wins.

THE RULE

A gap only shows up if it is **wider than about half your distance from the wall.** At 20 inches out you would need a gap nearly a foot across. **Want it to notice doorways? Drive closer.**

THE OTHER SENSOR

Infrared: it shines and looks.



It sends light

Infrared — just past red, where your eyes stop. **Same signal as a TV remote.**

It measures brightness

Not time. **How much came back.** Light spreads out, so less returns from further away.

You get 0 to 100

Not centimeters — it has no way to work those out. Useful to about **50–70 cm**.

THE WHOLE DIFFERENCE

The ultrasonic **times a journey.** The infrared **judges a brightness.** Everything else follows from that.

TWO WAYS TO SEE WITHOUT TOUCHING

Neither is better. They fail at different things.

A cushion

Ultrasonic: **nothing at all.** Infrared: **sees it clearly.** Soft, but pale.

A black card and a white card

Ultrasonic: **identical.** Infrared: **completely different.**

A wall turned 30°

Both fail. **Infrared does not fix everything.**

So which do you put on the robot?

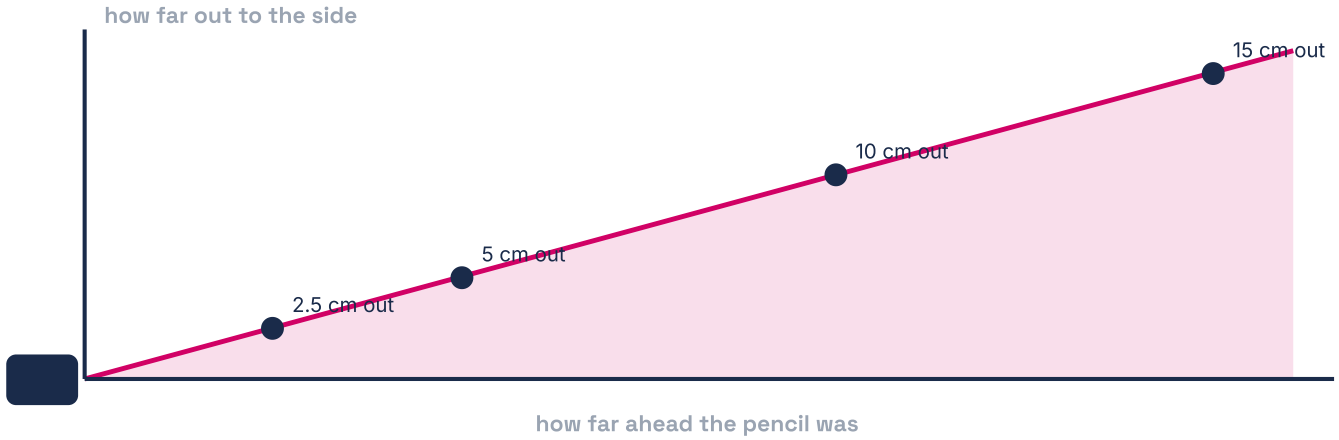
Wrong question. **What are you trying to see, and what is it made of?**

THE RULE UNDERNEATH BOTH OF THEM

Every sensor needs **one** thing from what it is looking at. Sound needs something **hard.** Light needs something that **reflects.** Anything without it is not badly seen — **it is not seen at all.** Their blind spots do not overlap, and that is why real robots carry more than one.

Draw the cone from your own measurements.

Stand a pencil out to the **side** of the sensor and walk it away until the robot first notices it. Mark that spot. Do it again from further out.

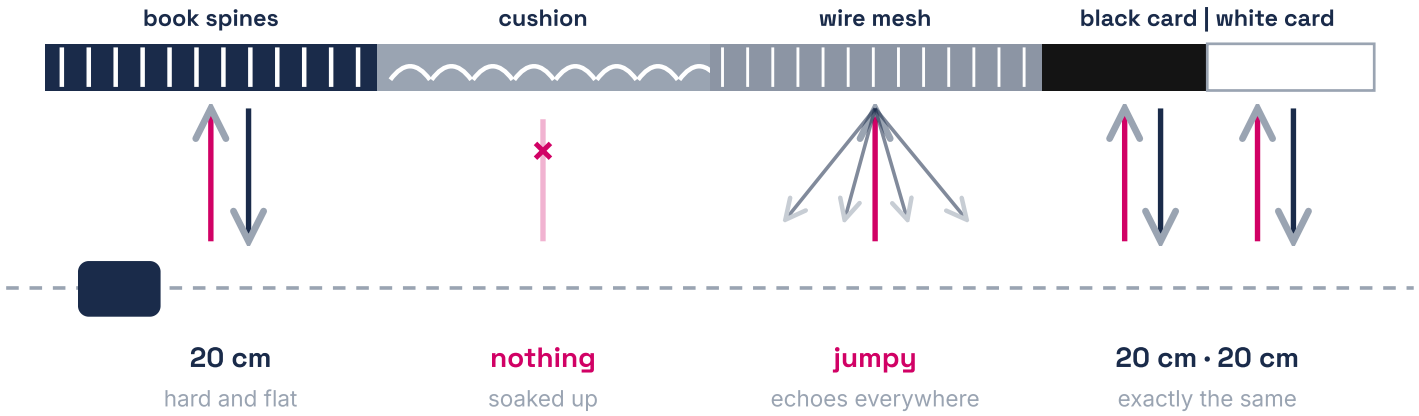


WHAT THE MARKS TELL YOU

They fall on a **straight line.** That is the whole finding — the beam does not bulge or fade, it opens out at a steady angle. **Twice as far ahead, twice as wide.** Everything inside your line is what the robot can see.

Some things send the sound back. Some do not.

Line two meters of wall with four surfaces. **Predict which ones the robot will lose**, then drive your wall follower along it and mark the floor where it actually fails.



THE ONE NOBODY PREDICTS

The black card and the white card read **exactly the same.** Color is something **light** does. A sound wave has no way of knowing what color anything is.

REFERENCE · KEEP THIS ONE OPEN

Every formula, in one place.

Six slides. **You do not have to remember any of them.** Look them up while you build — that is what engineers do.

★ **The only one worth knowing by heart**

speed = distance ÷ time

EVERYTHING ELSE

The rest is **one multiply** or **one divide**. Nothing on the next six slides is harder than that.

How far did it go?

Circumference

cm

π × across the middle

How far the wheel rolls in **one turn**. Your wheel is 56 mm across.

Distance

cm

wheel turns × 17.6

Count the turns, multiply. That is how far you went.

TRY IT

π × 56 mm = **17.6 cm** · 3 turns = 3 × 17.6 = **52.8 cm**

Gears change how far one motor turn takes you.

Gear ratio

wheel gear ÷ motor gear

Always **the gear on the wheel, divided by the gear on the motor** — that way it works whether you geared for push or for speed. A gear in the *middle* flips the direction but does not change the ratio.

Distance with gears

cm

17.6 ÷ ratio

How far **one motor turn** carries you once gears are in the way.

Working it backwards

17.6 ÷ distance you measured

Measure how far it went, and you can work out **what gears are inside.**

TRY IT

Geared for push: wheel 36 ÷ motor 12 = **3** → 17.6 ÷ 3 = **5.9 cm** a turn
Geared for speed: wheel 12 ÷ motor 36 = **1/3** → 17.6 ÷ 1/3 = **52.8 cm** a turn

Speed is distance ÷ time.

Speed

cm/s

distance ÷ time

Nothing on your robot measures speed. **You work it out** from a distance and a stopwatch.

Using the sensor

cm/s

(first reading − second reading) ÷ seconds

Two readings, a known time apart. **That is a speedometer.**

TRY IT

30 cm in 4 seconds → 30 ÷ 4 = **7.5 cm/s** · same as Week 3

How the sensor knows.

Speed of sound

m/s

343 m every second

In a warm room. **Slower when it is cold.** A slower echo takes longer, so a cold gym makes every robot read a little **far**.

There and back

cm

distance × 2

The sound goes to the wall **and comes back.** That is why the robot halves it.

What the robot does

cm

(343 × time) ÷ 2

It times the echo, then does the sum. **It never measures a distance.**

TRY IT

Wall 30 cm away → sound travels 60 cm → 0.60 ÷ 343 = **0.0017 seconds**

You can have push or speed. Not both.

Turning force

push × how far out you push

Push at **twice the distance from the middle** and you get twice the turning. It is why a long wrench undoes a tight bolt.

Power

push × speed

Your motor has **a fixed amount.** Gears move it between push and speed. **They never make more.**

Fast or strong. Never both.

TRY IT

3× the speed → about **1/3 the push.** Multiply them back together and you get the same number.

Can you trust your numbers?

Average

cm

add them up ÷ how many

Your best single answer from several tries.

Spread

cm

biggest − smallest

How much your tries disagreed. **Often more interesting than the average.**

Off by

cm

your average − the tape measure

If every try is low, **averaging will not save you.** Something is wrong.

TRY IT

29.4, 29.6, 29.5 · tape says 30 → average **29.5**, spread **0.2**, off by **0.5 cm**